

Dynamics of Computation in the Brain - Exercise 4

Due July 11, 2024

1 Analytical Exercise: Ring model with a complex input

Consider a network consisting of N neurons, arranged around a ring equally spaced between 0 and 2π , so that the i th neuron is labeled by an angle $\theta_i = \frac{2\pi i}{N}$ for $i = 0, \dots, N-1$. The connectivity is given by $\tilde{J}_{ij} = \frac{1}{N}\tilde{J}(\theta_i - \theta_j)$ with

$$\tilde{J}(\theta) = \begin{cases} J & |\theta| < \frac{\pi}{2} \text{ or } |\theta| > \frac{3\pi}{2} \\ 0 & \text{otherwise} \end{cases}, \text{ and the activities evolve according to}$$

$$\tau \frac{dr_i(t)}{dt} = -r_i(t) + \sum J_{ij}r_j(t) + h_i$$

where we will assume $\tau = 1[\text{sec}]$ for simplicity.

This time, the input to the system will be:

$$h_i = \begin{cases} 2 & |\pi - \theta_i| > \frac{\pi}{2} \\ 0 & \text{otherwise} \end{cases}$$

Note that much of the following was done in class.

1. Write the above equation in its continuous form, by taking $N \rightarrow \infty$.
2. Write both $r(\theta)$ and $h(\theta)$ as a Fourier series, writing $r(\theta)$ in terms of the unknowns $a_n(t), b_n(t)$. Solve the equations such that we will get uncoupled dynamical equations for $a_n(t), b_n(t)$.
For calculating the Fourier series for $h(\theta)$, you might use the fact the Fourier series of a very similar function was already calculated in Exercise 2.
3. What is the fixed point of the dynamics? That is, what are the values of a_n^*, b_n^* ?
4. What are the values of J for which this fixed point is stable?

2 Numerical Exercise: The ring model

2.1 Simulating the ring model

In what follows, we will simulate the above dynamics for $N = 1001$ and use $J = 1$ for all simulations¹.

1. Generate the connectivity matrix of the dynamics, $\tilde{J}_{ij}$. It might be useful to read about creating circulant matrices in python and matlab.
2. Simulate the dynamics of the network for $t = 1000[\text{sec}]$. How long did it take the simulation to run? In python, you can use the function `time.time()` to answer this question, while in Matlab this is done using the `tic()` and `toc()` functions.

Note: You will not be graded for efficient running time, but we will use it to let you know if you can run your code faster for the next exercise. Sometimes, by changing a small number of code lines one can improve the

¹We chose an odd number for N to avoid confusion regarding the connectivity in the finite case. In this case, each neuron is connected to the same number of neurons from both sides. However, the same analysis will work for an even N as well.

code's running time by a factor of 10 or more. In addition, it's important to know how to evaluate your codes running time.

3. Plot the activation pattern on the ring for $t = 1$ [sec], $t = 25$ [sec], $t = 50$ [sec], $t = 200$ [sec] and $t = 600$ [sec]. That is, plot the firing rate $r_i(t)$ as a function of the location on the ring, θ_i . In the same figure, plot the input to the system h_i . Comment on the differences between the different graphs, and about what happens to the dynamics as it progresses in time.

2.2 Studying the effect of J

Here, we will use the mean firing rate across all neurons on the ring as a measure to when the dynamics have reached the fixed point.

1. For $J = 1$, $J = 1.5$, $J = 1.9$ and $J = 2.1$ plot the average firing rate across the ring as a function of time. What can we learn from these graphs regarding the stability of the network and the time it takes it to reach the fixed point?
2. For a general function $f(x)$, what is the definition of a_0 , the zeroth Fourier component? What is the relation between a_0 and the average value of the function $f(x)$?
3. Use the dynamical equation of $\dot{a}_0$ derived in the first part of this exercise to explain the behavior of the graphs - both in term of the time it takes the mean to reach its steady state, and in terms of its final value.
4. Focus on the case of $J = 2.1$:
 - (a) According to the first part of this exercise, which coefficients a_n, b_n will have stable dynamics near the fixed point for this value of J ?
 - (b) Plot the firing rate $r_i(t)$ as a function of the location on the ring θ_i for several timepoints along the dynamics. How does the shape of the activation pattern throughout time reflects your answer to the previous question?

3 Qualitative Observations

1. When solving this model analytically, we used $N \rightarrow \infty$. Given the numerical results, was this approximation justified? What aspects of the dynamics are the same, and what aspects are different? How can we determine whether $N = 1001$ is "large enough" to be treated as "infinity"? If you want, you can run another simulation with a very small number of neurons to get some intuitions regarding the effect of the size of N .
2. The dynamics are determined both by the external input and the connectivity profile $\tilde{J}_{ij}$. How does the shape of the external input affect the dynamics? How does the connectivity profile $\tilde{J}_{ij}$ affect it? Relate to the stability of the dynamics, its fixed point and the time it takes the dynamics to converge to it.